0580_w24_ms_42_Q7

Question	Answer	Marks	Partial Marks	Guidance
7(a)(i)	180 – 60 – 39 [= 81]	**1**		Must show subtraction Do not accept 180 – 99 without 60 + 39 = 99 first
7(a)(ii)	147 or 147.1…	**3**	**M2** for $\frac{129\sin(81)}{\sin 60}$ oe or **M1** for $\frac{\sin(81)}{CD} = \frac{\sin 60}{129}$ oe	Longer methods possible M2 for *CD* explicit M1 for *CD* implicit
7(a)(iii)	[cos =] $\frac{85^2 + 129^2 - 72^2}{2 \times 85 \times 129}$	**M2**	**M1** for $72^2 = 85^2 + 129^2 - 2 \times 85 \times 129 \cos ABD$	See AG for alt method
	31.58…	**A2**	**A1** for 0.851 to 0.852 or $\frac{9341}{10965}$ or equivalent fraction	After M1 answer 31.58… implies M2A2 After M1, A1 implies M2 A1 for e.g. $\frac{72.4[1...]}{85}$
7(a)(iv)	44.5 or 44.51 to 44.54	**3**	**M2** for implicit correct method e.g. $\frac{d}{85} = \sin 31.6$ oe or **M1** for recognition that the line from *A* is perpendicular to *BD*	For M2, accept more accurate value for the angle. M2 for $\frac{1}{2} \times 85 \times 129 \times \sin 31.6 = \frac{1}{2} \times 129 \times h$ For info if angle *BDA* used then *BDA* = 38.2 May be on diagram e.g. right-angle symbol, accept e.g. $\frac{1}{2} \times 129 \times h$
7(a)(v)	247 or 247.4…	**2**	**M1** for 180 + (180 – 81 – 31.6) oe or for ∠*NBC* = 180 – 81 – 31.6 oe or for ∠*NCB* = 81 + 31.6 oe	For M1, accept more accurate value for 31.6 *N* = North line at *B* [67.4] *N* = North line at *C* [112.6] For M1 angles must be identified, may be on diagram.

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Question	Answer	Marks	Partial Marks	Guidance
7(a)(vi)	972 or 973	**4**	**M1** for [$\triangle ABD$] $\frac{1}{2} \times 85 \times 129\sin 31.6$ oe or $\frac{1}{2} \times 129 \times$ *their* 44.5 oe **M1** for [$\triangle BCD$] $\frac{1}{2} \times 129 \times$ *their* $147 \times \sin 39$ oe **M1** for $\frac{\textit{their}\text{ total area}}{10000} \times 1100$	Condone non-integer answer, 972. … Accept more accurate values for 31.6, 44.5 and 147. $\triangle ABD$ = 2870 to 2873 If other angles used, angle BAD = 110.[2…], angle BDA = 38.2 $\triangle BCD$ = 5966 to 5972 M1M1 implied by 8840 or 8850 or 8836 to 8845 If other length used $BC \approx 93.7$ If additional incorrect areas are added to make the total area then award M1M0 Allow use of what they think is *their* total area
7(b)	126 nfww	**3**	**M2** for $\frac{9400+50}{70 \text{ to } 80}$ or $\frac{9400 \text{ to } 9500}{80-5}$ or **M1** for 9350 or 9450 or 75 or 85 seen	For M1 accept 9400 – 50 or 9400 + 50 or 80 – 5 or 80 + 5 seen